

SEBASTIAN CORREA-GALLEGO

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Research Interests

I am interested in how order emerges from parts that were previously autonomous, and in whether that process follows rules or depends on history — a question I find most tractable in microbial community assembly, where the same organisms may converge on one reproducible state or settle into any of several, depending on who arrives first. I ask whether that outcome can be anticipated from the organisms themselves, from how cells allocate their finite resources and how selection reshapes those strategies while a community assembles, and I want to follow the question upward, toward the thresholds at which cooperating parts cease to be separable. My approach is experimental, combining cultivation, growth physiology, and quantitative analysis of community structure.

Education

Universidad EAFIT, Medellín, Colombia

2022–2026

B.Sc. in Biology, conferred June 2026

GPA: 4.44 / 5.00

Relevant coursework: Ecology; Evolution; Microbiology; Biostatistics and Experimental Design; Dynamic Systems; Fundamentals of Computational Biology; Data Science in Life Sciences; Experimental Biology.

Research Experience

Visiting Student Intern

Aug 2025–Jan 2026

[The Evolution of Cells, Symbionts and Organelles Lab](#), Purdue University

Advisor: Sergio A. Muñoz-Gómez, Ph.D.

Direct supervision: Shahed U. A. Shazib, Ph.D.

- Investigated how non-conventional osmotrophic yeasts (*K. marxianus* and *V. polyspora*) reorganize proteome investment under carbon limitation, within the project *Proteome Allocation Rules in Osmotrophic Eukaryotes*, addressing how cells partition finite resources between growth and metabolic capacity.
- Generated high-resolution growth curves across defined carbon regimes for two organisms with three replicates per condition, establishing the growth phenotypes underlying the downstream analysis.
- Contributed to the proteomics workflow from sample preparation through integrative analysis linking molecular allocation to cellular physiology, and estimated cell volume from microscopy images by manual measurement of both axes across approximately 2,100 cells, with all analysis conducted in R.
- Tested and calibrated an automated continuous-culture system (chemostat and turbidostat modes) to support controlled cultivation for the lab's steady-state and experimental-evolution work.

Undergraduate Thesis Researcher

2024–2026

Universidad EAFIT

Advisor: Nicolás Pinel Peláez, Ph.D.

Thesis: [Cultivable Microbial Community Structure Along a Light Gradient in a Tropical Volcaniclastic Cave](#)

- Tested whether light availability, as a proxy for energy input, structures the cultivable microbial community of the Organal San Antonio — a tropical volcaniclastic cave at ~2350 m a.s.l. in Támesis, Colombia — establishing a spatially resolved cultivable baseline for an organal-type pseudokarstic system.
- Designed and executed three field campaigns, advancing from exploratory sampling to a categorical light–dark contrast and finally to a gradient design with three qualitatively characterized sectors and five replicates each, adapting the cultivation setup to available resources and seeking external funding through two competitive schemes.

- Found an approximately sixfold reduction in cultivable density in the aphotic sector, which supported a community compositionally distinct from the photic sectors, with roughly a quarter of the Transition assemblage restricted to that zone alone and composition partitioning strongly with zonation (PERMANOVA $R^2 = 0.57$); the comparison required a morphotype crosswalk joining per-zone libraries into comparable operational units, whose resolution limit is stated explicitly in the thesis.

Manuscripts

Proteome Allocation Rules in Osmotrophic Eukaryotes

Coauthored manuscript in preparation

The Evolution of Cells, Symbionts and Organelles Lab, Purdue University

Conferences and Presentations

Cultivable Microbial Community Structure Along a Light Gradient in a Tropical Volcaniclastic Cave

Oral Presentation

May 2026

V Simposio de Biología Divulgativa, Universidad EAFIT, Medellín, Colombia

Proteome Allocation Rules in Osmotrophic Eukaryotes

Oral Presentation

Jan 2026

UREP-C and UREP-LA Final Symposium, Purdue University, West Lafayette, IN, USA

Characterization of Cave-Dwelling Phototrophic Microbial Communities from the Northwestern Andes

Poster Presentation

Nov 2024

II Simposio de Biología Divulgativa, Universidad EAFIT, Medellín, Colombia

Positioning Microorganisms as Key Actors in the Maintenance of Andean Forests through Science Communication

Poster Presentation

May 2024

XXIII Encuentro Departamental de Semilleros de Investigación, RedCOLSI, Medellín, Colombia

Exploration of Organel Ecosystems in the Organel San Antonio (Támesis, Antioquia) and Identification of the Psychrophilic Photosynthetic Microorganisms Inhabiting Them

Poster Presentation

May 2023

XXII Encuentro Departamental de Semilleros de Investigación, RedCOLSI, Medellín, Colombia

Exploration of the Thermoacidophilic Microorganisms Inhabiting the Aguas Calientes Hot Spring

Poster Presentation

Sep 2022

Feria de Semilleros de Investigación, Universidad EAFIT, Medellín, Colombia

Fellowships and Grants

Simons Graduate Fellowships in Ecology and Evolution

2026

Simons Foundation

Application submitted June 2026; *under review*, notification expected October 2026.

Research Internship Sponsorship

2025

Fundación Fraternidad Medellín

Full funding for a six-month research internship at Purdue University (UREP-C Program).

Undergraduate Scholarship

2022–2026

Fundación Fraternidad Medellín

Full tuition scholarship for the B.Sc. in Biology, awarded on academic merit.

Honors and Awards

Second Prize, Concurso de Relatos Científicos

2024, 2025

Biology Program, Universidad EAFIT

Faculty-judged science writing competition. Second place in two consecutive editions: I Concurso, Eastern Cordillera category, for *La Clínica de la Sombra* (Nov 2024); II Concurso, Western Cordillera category, for *Ethos de una Palma Andina* (May 2025).

Second Prize, Poster Presentation
II Simposio de Biología Divulgativa, Universidad EAFIT

Nov 2024

Academic Service

Student Coordinator, Research Group on Microbiology and Astrobiology (SIAB) *Jan 2024–Jun 2025*
Universidad EAFIT, affiliated with the Research Group on Geosciences and Biodiversity (GEBI)
Led a student research group on microbiology, astrobiology, and extremophiles, working collaboratively with members to learn and apply microbiological techniques and explore the diversity of these organisms. Member since January 2022, initially through the University's social-service requirement.

Teaching, Mentoring and Outreach

Volunteer Science Instructor and Materials Developer *2025–2026*
Institución Educativa de Desarrollo Rural Miguel Valencia, Jardín, Colombia

- Completed two 30-hour social-service projects (60 hours total) in support of the lower-secondary Natural Sciences course at a rural public school, in a mixed in-person and remote format.
- Accompanied the classroom teacher during instruction and developed digital teaching materials on the origin of the universe, microbiology, evolution, and biological diversity.

Project Monitor, Science-Outreach Initiative on Microorganisms and Andean Forests *2024*
Research Group on Microbiology and Astrobiology (SIAB), Universidad EAFIT

- Assumed day-to-day leadership of the initiative during its final six months, working with a designer to produce science-communication material on the role of microorganisms in the maintenance of Andean forests for the El Globo nature reserve (Támesis, Colombia).

Logistics Volunteer, Universo Semilleros Research Fair *Aug 2024*
Universidad EAFIT
Institutional research-presentation event organized by the Vice-Rectorate for Science, Technology and Innovation.

Training and Professional Development

Stanford Biology Preview Program (BPP) *Sep 2026*
Department of Biology, Stanford University
Selected participant, 2026 cohort.

Understanding the Undergraduate Research Experience (GS 29501) *Fall 2025*
Purdue University

Responsible Conduct of Research (RCR) Training *Aug 2025*
CITI Program, Credential ID 71597371, valid through Aug 2029

Agricultural Production Technician *2019–2021*
Servicio Nacional de Aprendizaje (SENA), Colombia

Workshops and short courses *2022–2023*
Perspectivas para un Impacto Sostenible (Institute of Management in Latin America, University of St. Gallen, 2023);
Aprendamos de Microbioma: Somos más que Células Humanas (Laboratorio Clínico Hematológico, 2022).

Technical Skills

Laboratory and field methods: microbial cultivation and isolation, continuous-culture operation (chemostat and turbidostat), growth-curve measurement, proteomics sample preparation, light and epifluorescence microscopy (MitoTracker and DAPI staining) with image-based cell measurement, environmental sampling, and ecological field records. Coursework-based bench training in molecular biology and biochemistry.

Quantitative and computational tools: R (multivariate statistics, ecological data analysis, visualization), QGIS, and $\text{\LaTeX}$; Python and Linux/bash (coursework-based).

Languages: Spanish (native); English (professional working proficiency).